HINTS & SOLUTIONS

Q.1 (C)

$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$

$\left(\dfrac{1-\alpha}{1+\alpha}\right)P \qquad \left(\dfrac{2\alpha}{1+\alpha}\right)P \qquad K_P = \dfrac{4\alpha^2}{1-\alpha^2}P$

$\alpha_1 = \dfrac{1}{3}$ at P_1 and $\alpha_2 = \dfrac{1}{2}$ at P_2

$$\left\{\frac{4\left(\frac{1}{3}\right)^2}{1-\left(\frac{1}{3}\right)^2}\right\}P_1 = \left\{\frac{4\times\left(\frac{1}{2}\right)^2}{1-\left(\frac{1}{2}\right)^2}\right\}P_2$$

$$\frac{P_1}{P_2} = \frac{\frac{8}{9}}{\frac{4}{9}}\times\frac{\frac{1}{3}}{\frac{3}{4}} = \frac{8}{3}$$

Q.2 (D)

$Cl_5(g) \rightleftharpoons PCl_3(g) + Cl_2$

t = 0 0.1 mol 0 0

$t = t_{eq}$ (0.1–x)mol x-mol x-mol

$$\therefore K_C = \frac{\frac{x}{5}\times\frac{x}{5}}{\frac{(0.1-x)}{5}} \Rightarrow 10 = \frac{x^2}{5(0.1-x)}$$

$\Rightarrow 50\,(0.1 - x) = x^2 \Rightarrow x^2 = 5 - 50\,x$

$\Rightarrow x^2 + 50\,x - 5 = 0 \quad \therefore x = 0.0998 \approx 0.1$

$\therefore n_{PCl_5}$ is negligible

$$\therefore P_{final} = \frac{(0.1+0.1)\times 0.08\times 600}{5}\,\text{atm} = 1.92\text{ atm}$$

Q.3 (D)

$$2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$$

$$a - ax \qquad ax \qquad \frac{ax}{2}$$

$$K_P = \frac{\left(\dfrac{ax}{a+\dfrac{ax}{2}}P\right)^2\left(\dfrac{ax/2}{a+\dfrac{ax}{2}}P\right)}{\left(\dfrac{a-ax}{a+\dfrac{ax}{2}}P\right)^2}$$

$$x = (2K_P/P)^{1/3}$$

Q.4 (D)

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

$$P \rightarrow \quad 1-\alpha_1 \qquad \alpha_1 \qquad \alpha_1$$

Total pressure $K_p = \dfrac{\left(\dfrac{\alpha_1}{1+\alpha_1}P\right)^2}{\left(\dfrac{1-\alpha_1}{1+\alpha_1}O\right)^1} = \dfrac{\alpha_1^2 P}{1-\alpha_1^2}$

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

$$1-\alpha_2 \qquad \alpha_2 \qquad \alpha_2$$

$$K_p = \frac{\alpha_2^2 P/2}{1-\alpha_2^2}$$

$$\frac{\alpha_1^2 P}{1-\alpha_1^2} = \frac{\alpha_2^2 P/2}{1-\alpha_2^2}$$

$$\alpha_2^2 = 2\alpha_1^2$$

$$\alpha_2 = \sqrt{2}\alpha_1$$

Q.5 (C)

$$X(g) \rightleftharpoons Y(g) + Z(g)$$

$$\left(P - \frac{6P}{7}\right) \qquad \frac{6P}{7} \qquad \frac{6P}{7}$$

$$K_p = \frac{\left(\dfrac{6P}{7}\right)\left(\dfrac{6P}{7}\right)}{\dfrac{P}{7}} = \frac{36P}{7}$$

Q.6 (A)

Q.7 (B)

$A.3H_2O(s) \rightleftharpoons A.H_2O(s) + 2H_2O(g)$ $\qquad K_P = 9\ atm^2$

x mole $\qquad K_P = P^2_{H_2O}$

PV = nRT

$3\ at \times 24.63 lt = n_{H_2O} \times 0.0821 \times 300$

$n_{H_2O} = 3$ mole

5 mole of H_2O (g) should be added.

Q.8 (A)

$Q = P_{NH_3} \times P_{H_2S} = 50\ atm^2$ as $Q < K_P$, the reaction should go in forward direction, but $NH_4HS(s)$ is not present in container $\therefore$ there will be no change in pressure]

Q.9 (B)

$$-\frac{d[A]}{dt} = \frac{1}{x}\frac{d[B]}{dt} = \frac{1}{y}\frac{d[C]}{dt}$$

$$-\frac{(2-3)}{t} = \frac{1}{x}\frac{(2-0)}{t} \Rightarrow x = 2$$

$$-\frac{(2-3)}{t} = \frac{1}{y}\frac{(1-0)}{t} \Rightarrow y = 1$$

$[A]_{eq} = 2,$ $\quad [B]_{eq} = 2,\ [C]_{eq} = 1$

$$\therefore K_C = \frac{[B]^x\ [C]^y}{[A]} = \frac{2^2 \times 1^1}{2} = 2$$

Q.10 (B)

Q.11 (D)

Q.12 (C)

Q.13 (A)

Q.14 (C

$K_P = P_{H_2O}$

K_P is temperature dependent only.

Q.15 (B)

$K_p = P^2$

$$\ln\frac{K_{P_2}}{K_{P_1}} = \frac{\Delta H}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\ln\frac{P_{400}^2}{P_{300}^2} = \frac{16.628\times10^3}{8.314}\left[\frac{1}{300} - \frac{1}{400}\right]$$

$$2\ln\frac{P_{400}}{P_{300}} = 2\times10^3\left[\frac{100}{300\times400}\right] = \frac{10}{12} = \frac{5}{6}\ \textbf{Ans.}$$

Q.16 (D)

	$N_2O_{4(g)}$	$\rightleftharpoons$	$2NO_{2(g)}$
Initial moles	1		0
At eq^m moles	$1-\alpha$		2α

$$K_P = \frac{(2\alpha)^2 P}{(1-\alpha^2)} = \frac{4\alpha^2 P}{(1-\alpha^2)}$$

$$\frac{4\times(0.1)^2\times3.3}{1-(0.1)^2} = \frac{4\times(0.2)^2\times P}{1-(0.2)^2}$$

$$\frac{3.3}{0.99} = \frac{4P}{0.96} \Rightarrow P = 0.8\text{ atm}\quad \textbf{Ans.}$$

Q.17 (B)

Balanced Chemical Equation is

$3I_2\,(s) + 6\,OH^-\,(aq) \rightleftharpoons 5I^-(aq) + IO^-_3\,(aq) + 3H_2O(l)$

$\Delta G° = 5\times(-50) + (-123.5) + 3\times(-233) - 0 - 6\times(-150) = -172.5$ kJ/ mol]

Q.18 (C)

$\Delta G° = -2.3\,RT\log k = -172.5\times10^3$ J/mol

$$\Rightarrow -2.3\times\frac{25}{3}\times300\log k = -172.5\times10^3$$

$\Rightarrow \log k = 30$

$k = 10^3$]

Q.19 (A)

Sol. $k = \dfrac{[I^-]^5[IO_3^-]}{[OH^-]^6} = 10^3$

$$\Rightarrow \frac{[0.1]^5[0.1]}{[OH^-]^6} = 10^{30}$$

$[OH^-] = 10^6$ M $\qquad$ pOH = 6 $\qquad \therefore$ pH = 8

Q.20 (C)

$CH_3COOH_{(aq)} + H_2O \rightleftharpoons CH_3COO^-_{(aq)} + H^+_{(aq)}$

t = 0	a moles		
t_{eq}	a – x	x	x

$$\therefore Q = \frac{x^2}{a-x} \times \frac{1}{V}$$

On adding water, V increases so decreases. So reaction will move in forward direction. NaOH will react with H^+. So $[H^+]$ decreases so reaction will move in forward direction. Addition of HCl will increase concentration of H^+ in solution. Reaction will shift in backward direction. Removal of CH_3COO^- will shift reaction in forward direction.

Q.21 (D)

	$3A_{(g)}$ +	$2B_{(g)}$ $\rightleftharpoons$	$C_{(g)}$ +	$5D_{(g)}$
t_{eq}	4 moles	5 moles	8 moles	6 moles
	2M	2.5 M	4 M	3 M
t'	0.4 M	0.5 M	0.8 M	0.6 M
t'_{eq}	0.4 – 3x	0.5 – 2x	0.8 + x	0.6 + 5x

$[C]_{new\ eq}$ = 0.8 + x which can not be greater than or equal to 4 M.

Q.22 (B)

$C_{(graphite)} \rightleftharpoons C_{(diamond)}$ $\qquad$ Endothermic reaction

at 500 K reaction will shift in forward direction.

$P < 1.5 \times 10^9$ Pa

[As $C_{(diamond)}$ is more dense]

Q.23 (B)

Q.24 (B)

Q.25 (A)

(i) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$t = 0$ a b

t_{eq} $a - x - y$ $b - 3x - 2y$ $2x$

$N_2(g) + 2H_2(g) \rightleftharpoons N_2H_4(g)$

$t = 0$ a b

t_{eq} $a - y - x$ $b - 2y - 2x$ y

Now, $P_{NH_3} = 2\,atm$

$\therefore \quad 2x = 2 \Rightarrow x = 1$

Also $P_{H_2} = 4\,atm$

$b - 3 - 2y = 7$ .....(i)

Total pressure = 14 atm

$a - x - y + 4 + 2 + y = 14$

$a - x = 8$

$a = 9$ atm

(as molar ratio of a & b = 9 : 13)

$b = 13$ atm

(i) $\Rightarrow 2y = b - 7 = 6$

$y = 3$ atm

$P_{N_2} = a - x - y = 9 - 1 - 3 = 5$ atm

$P_{H_2} = 4$ atm

$P_{NH_3} = 2$ atm

$P_{N_2H_4} = 3$ atm

$$K_{P_1} = \frac{2 \times 2}{4 \times 4 \times 4} = \frac{1}{80}\ atm^{-2}$$

(ii) $$K_{P_2} = \frac{3}{5 \times 4 \times 4} = \frac{3}{80}\ atm^{-2}$$

$$\therefore \frac{1}{K_{P_2}} = \frac{80}{3}\ atm^2$$

(iii) $P_{N_2} = 5$ atm

Q.26 (C)

Q.27 (B) (D)

$K_{eq} = Ae^{-\Delta H/RT}$

Taking log on both the sides

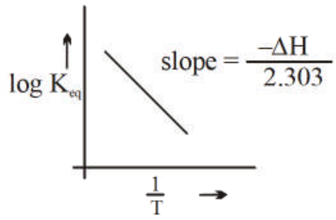
$$\log K_{eq} = \log A \frac{-\Delta H}{2.303R} \cdot \frac{1}{T}$$

$Y = +C - m x$

thus $\Delta H = +ve$

So the reaction is endothermic on increasing temperature equilibrium shift in forward direction concentration of Cl_2 increases.

On removing F_2 (g) equilibrium shift [B] inforward direction concentration of Cl_2 increases [D]]



Q.28 (B) (C)

$A_{(g)} \rightleftharpoons 3B_{(g)} + C_{(g)}$ $\Delta H = +ve$ at 400K

P_{total} at $eq^m = 1$ bar

⇒ on increasing T, reaction moves in forward direction.

⇒ on introducing inert gas at const. T, total pressure increases but partial pressure remains the same so no effect at eq^m

⇒ at 400K $K_P \neq 1$ (since $P_{total} = 1$ bar)

$\Delta G°_{400K} = -RT \ln K_P \neq 0$

⇒ If V is increased, partial pressure of each species would be less than initial value.]

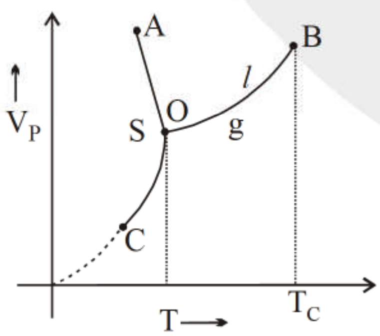
Q.29 (C) (D)

Q.30 (A) (B)

Q.31 (B) (D)

Q.32 (A) (B) (C) (D)

For water



O is triple point and B is critical point.

(A) At absolute zero vapour pressure of solid will be zero.

(B,C) Beyond triple point it will get convert into other phases and liquid phase has minimum pressure at this point.

(D) Beyond critical point phase of liquid will get changed.

Q.33 (A) (C) (D)

(a) $[A]+[B]+[C]=1$ mole/L since for every reaction $\Delta n_g=0$

(b) $\frac{[C]}{[B]}=0.4 \Rightarrow [C]=0.4\,[B]$

$\frac{[A]}{[C]}=0.6 \qquad [A]=0.6\,[C]$

$[A]+[B]+[C]=1$

$0.6[C]+\frac{[C]}{[0.4]}+[C]=1$

$[C]\left\{0.6+\frac{1}{0.4}+1\right\}=1$

$[C]=\frac{4}{16.4}=\left[\frac{1}{4.1}\right] \qquad \Rightarrow \qquad \frac{[B]}{[A]}=\left[\frac{1}{0.24}\right]=K_1$

Q.34 (A) R (B) Q (C) S (D) P

Q.35 (A) P, Q, R, S ; (B) Q, R, S ; (C) P, Q, R, S, T ; (D) Q, R, **S**

Q.36 0250

$C(s) \rightleftharpoons A+B$

5–x y x

$A+D \rightleftharpoons E(s)$

x 5

y 5–x x

$xy=5\times10^{-11}$

$y\,(5-x)=10^{-10}$

$\frac{5\times10^{-11}(5-x)}{x}=10^{-10}$

$2.5-0.5\,x=x$

$x=\frac{5}{3}$

Q.37 2.7 g / lit

Q.38 4